

CSC 450, Senior Research Literature Review

1. Write a **literature review**. The literature review should be roughly two double-spaced pages (not including references). The literature review contains background information about a computer science topic, the significance of the topic, discusses related work, and would support the research project you are pursuing. Your literature review should have 4-5 sections: Background, Significance, Related Work, Acknowledgments (Optional) and References. Your review should be written appropriately for a reader who is a senior undergraduate student in computer science. **The literature review must have at least 8 references, and at least 3 references must be research articles that are included in the *Related Work* section of your literature review.** *You must also submit an export of your Zotero library with your literature review.* The literature review will serve as a basis for your proposal, will provide introductory material and discussion points for your final paper, and may also generate ideas for your project (such as for its methodology). However, the literature review does not describe your specific project, but instead describes background and work that has been done in related areas. Note that *related* work includes work related to any aspect of your project, whether it relates to the general topic, the specific question, or related methodologies.

You must submit this assignment through Blackboard by the due date to receive credit. Late submissions will not be accepted. This assignment will be evaluated using the rubric on the following page:

Total:

Background information (this is the first section of the Lit Review) / 30 points

- Background is presented in a clear and organized manner
- Background is described to ensure that the reader can understand the general topic; key terms and concepts are defined.

Significance (this is the second section of the Lit Review) / 30 points

- Significance is presented in a clear and organized manner
- Significance includes why the research topic is important for its field and society as a whole
 - How does society benefit from studying this topic? Who, specifically, does this topic benefit?

Related Work (this is the third section of the Lit Review) / 40 points

- Related work is presented in a clear, organized, and critical manner
- The methodology, major findings, significances, weaknesses, and limitations of at least 3 research articles are discussed.
- Related work includes articles with similar research questions, objectives, methods, or datasets.

Additional Lit Review Requirements

- **Title:** The paper has a descriptive and specific title. For example, “Machine Learning” is too broad a topic, but “Machine Learning Methods” or “Machine Learning Applications” would be acceptable. [**5 point deduction if not met**]
- The paper does *not* describe your specific project (and therefore does not contain “I” or “we”) [**10 point deduction if not met**]
- **References:** The paper contains at least 8 references that are cited in the manuscript and included in the *References* section using IEEE format. References need to be cited in the body of the lit review, and not just appear at the end. You must use Zotero to manage your references, and submit your exported Zotero library with your Literature Review. [**paper is not acceptable if it does not contain any citations or any references; up to 20 point deduction if Zotero library is not submitted**]

- **Format:** The paper is approximately 2 pages (excluding references) and double-spaced with 1" margins. All paragraphs except for the first in a section are indented. Arial (font size 11) or Times New Roman (font size 12) are used throughout. The paper includes a title in bold, followed by your name. The title may be one font size larger. At the end of the paper is a *References* section. The word *References* is bold, and the font size for the references section is 1 size less than the rest of the paper. The paper includes the following sections with headers in bold: Background, Significance, and Related Work. [**up to 10 point deduction if not met**] _____
- **Spelling and Grammar:** The paper is well-written and contains few, if any spelling mistakes or grammatical mistakes that detract from the reader's ability to understand the paper [**up to 20 point deduction if not met**] _____
- **Use of AI.** We will follow the IEEE policy on Author Guidelines for AI-generated text (<https://open.ieee.org/author-guidelines-for-artificial-intelligence-ai-generated-text/>), which requires an acknowledgement if AI is used; and the Chicago Manual of Style (<https://www.chicagomanualofstyle.org/qanda/data/faq/topics/Documentation/faq0422.html>), which requires footnotes that should include specific prompts. If you use AI for any purpose, you must include an Acknowledgements section which documents this.